GGG-W-001405
January 31, 1968

INTERIM FEDERAL SPECIFICATION

WRENCH, BOX (RATCHETING)

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D.C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers box end wrenches having a ratcheting mechanism in each head.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. The wrenches shall be of one type.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424 - Hand Tools, Packaging of.

Military Standards:

MIL-STD-8 Dimensioning and Tolerancing.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of wrenches otherwise conforming to the requirements of this specification.

FSC 5120

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:43Z
Check the source to verify that this is the current version before use.

3.2 Material. The wrenches shall be steel.

3.3 Marking. Wrenches shall be marked in a permanent manner with the manufacturer's name or trademark and with the nominal tool size.

3.4 Dimensions. All dimensions shall be interpreted as specified in MIL-STD-8.

3.5 Nut openings. Nut openings shall be of the proper design and size for accepting a plug gage conforming to the lower limits of table I and rejecting a similar gage conforming to the upper limits. The gages shall be hexagonal in section.

3.6 Test loads. When tested as specified in 4.5, the wrenches shall withstand the test loads of table I without breakage, distortion of the wrench opening which would permit acceptance of the maximum plug gage specified in 3.5, or any visible permanent distortion.

3.7 Coating and finish. The top and bottom of the wrench handle shall be either cadmium, nickel, or chromium plated and the two sides shall be either painted or enameled; or plated the same as on the top and bottom. The surface shall be free from burrs and cracks.

3.8 Hardness. Each socket and pawl shall be either through-hardened or case-hardened (testing with a file to determine that these parts have been hardened will be sufficient).

3.9 Fabrication. The wrenches shall be fabricated to provide a comfortable handgrip. Wrench bodies or handles shall be of laminated construction. Headed-studs, or rivets shall be neatly set.

3.10 Design. The wrenches shall be of double head design, of either 6 or 12-point style, and suitable for ratcheting. The reversing of the direction of the ratcheting operation shall be accomplished either by a shifting device or by the operator turning the wrench over. The wrenches shall be similar to figure 1 and conform to table I.

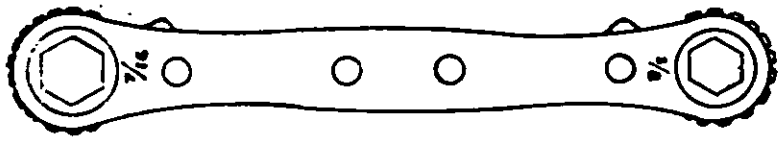


FIGURE 1. Wrench, Box (Ratcheting).

TABLE I. Wrench, Box (Ratcheting)

Wrench opening		Opening Dimensions				Outside diameter of head (max.)		Thickness of each head (max.)	Overall length (min.)	Test load	
										Small head	large head
Small head	Large head	Small head		Large head		Small head	Large head			(min.)	(min.)
Inch	Inch	Inch Min.	Inch Max.	Inch Min.	Inch Max.	Inches	Inches	Inch	Inches	Inch-pounds	Inch-pounds
1/4	5/16	.252	.257	.316	.322	3/4	13/16	7/16	3-3/4	144	180
3/8	7/16	.378	.384	.440	.446	15/16	1-1/16	15/32	5-1/4	360	420
1/2	9/16	.504	.510	.566	.573	1-5/32	1-7/16	1/2	6-1/2	600	720
5/8	11/16	.629	.636	.692	.699	1-3/8	1-9/16	1/2	7-3/4	900	930
5/8	3/4	.629	.636	.755	.763	1-3/8	1-9/16	1/2	7-3/4	900	960
11/16	7/8	.692	.699	.880	.888	1-9/16	1-9/16	1/2	8-5/8	930	1060
3/4	7/8	.755	.763	.880	.888	1-9/16	1-9/16	1/2	8-5/8	960	1060
13/16	15/16	.818	.826	.944	.953	1-9/16	1-5/8	3/4	8-5/8	1000	1100

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:43Z
Check the source to verify that this is the current version before use.

3.11 Workmanship. Workmanship shall be first class in all respects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies conform to prescribed requirements.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table II.

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table III.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.5.

TABLE II. Sampling Data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed In Terms Of
Visual examination	One each	II	1.5	Percent defective
Dimensional examination	One each	II	1.5	Percent defective
Testing	One each	S.4	1.5	Percent defective
Preparation for delivery	One each	II	4.0	Defects per hundred units

TABLE III. Classification of defects

Examine	Defect
Design	Not as specified.
Material	Not as specified.
Finish	Not as specified.
Coating	Not as specified.
Construction	Not as specified.
Workmanship	Not as specified.
Marking	Not as specified.

4.5 Test procedures.

4.5.1 Hardness. Tools shall be tested to determine compliance with the hardness requirements specified in 3.8.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:43Z
Check the source to verify that this is the currentversion before use.

4.5.2 Load test. Load tests shall be conducted on the sample tools to determine conformance with the applicable test load requirements specified in section 3. The loads shall be applied either with suitable torque producing machines or by means of a suitable lever with dead weights or testing machine. The load shall be applied for 10 seconds and then removed. Care shall be exercised to maintain accuracy in applying the load and in measuring the effective lever arm through which the load acts. The effective lever arm, when the load is vertical and lever is employed, is the shortest distance in inches, between two vertical lines passing, respectively, through the point of load application and the center of the mandrel fitting the wrench opening. The specified test load shall be the product of the effective lever arm, in inches; and the applied load, in pounds. A hexagon steel mandrel hardened to a Rockwell C 55 minimum shall be used.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation, and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A. Preservation and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 Level C. Cleaning, preservation, and packaging shall be in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Wrenches, packaged as specified (see 5.1), shall be packed in accordance with the level A requirements of MIL-H-15424.

5.2.2 Level B. Wrenches, packaged as specified (see 6.1), shall be packed in accordance with the level B requirements of MIL-H-15424.

5.2.3 Level C. Wrenches, packaged as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations as applicable to the mode of transportation.

5.3 Marking

5.3.1 Military agencies. In addition to any special marking required by the contract or order, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special marking required by the contract or order, (see 6.1), interior and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of tool required (see table I).
(c) Applicable levels of preservation, packaging, and packing (see section 5).
(d) Marking, if different (see 5.3).

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:43Z
Check the source to verify that this is the currentversion before use.